

Lecture 4

Hilbert spaces: A brief overview

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Fundamentos de Computação Quântica
Universidade do Minho
2025-2026

Recall: What is a qubit?

A **qubit** is a quantum system represented as a **superposition**, i.e. a linear combination of normalised and mutually orthogonal quantum states labelled by $|0\rangle$ and $|1\rangle$ with **complex** coefficients:

$$|v\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

When state $|v\rangle$ is **measured** (i.e. **observed**) one of the two basic states $|0\rangle, |1\rangle$ is returned with probability

$$|\alpha|^2 \quad \text{and} \quad |\beta|^2$$

respectively.

What is a qubit?

Being probabilities, the norm squared of the coefficients must satisfy

$$|\alpha|^2 + |\beta|^2 = 1$$

which enforces quantum states to be represented by **unit** vectors.

In practice, a qubit is a microscopic system, such as an atom, a nuclear spin, or a polarised photon.

The state space of a qubit

Global phases are useless

Unit vectors equivalent up to multiplication by a complex number of modulus one, i.e. a **phase factor** $e^{i\theta}$, represent the **same** state.

Let $|v\rangle = \alpha|u\rangle + \beta|u'\rangle$ and $e^{i\theta}|v\rangle = e^{i\theta}\alpha|u\rangle + e^{i\theta}\beta|u'\rangle$

$$|e^{i\theta}\alpha|^2 = (\overline{e^{i\theta}\alpha})(e^{i\theta}\alpha) = (e^{-i\theta}\overline{\alpha})(e^{i\theta}\alpha) = \overline{\alpha}\alpha = |\alpha|^2$$

and similarly for β .

As the probabilities $|\alpha|^2$ and $|\beta|^2$ are the **only** measurable quantities, the **global phase has no physical meaning**.

Representation redundancy

A qubit is not represented by a vector, but by **the subspace it generates**.

The state space of a qubit

Relative phase

It is a measure of the angle between the two complex numbers.
Thus, it **cannot be discarded!**

Those are different states

$$\frac{1}{\sqrt{2}}(|u\rangle + |u'\rangle) \quad \frac{1}{\sqrt{2}}(|u\rangle - |u'\rangle) \quad \frac{1}{\sqrt{2}}(e^{i\theta}|u\rangle + |u'\rangle)$$

...

Summing up

The State (or representation) Postulate

The state space of a quantum system is described by a unit vector in a Hilbert space up to a global phase

- In practice, with finite resources, one cannot distinguish between a **continuous** state space from a **discrete** one with arbitrarily small minimum spacing between adjacent locations.
- One may, then, restrict to **finite-dimensional** Hilbert spaces.

The underlying maths is that of Hilbert spaces.

A Language: Dirac's notation

Dirac's bra/ket notation provides a handy (easy to calculate with) way to represent elements of **vector space over the complex field**

$|u\rangle$ A **ket** stands for a vector in an Hilbert space V . In $\mathbb{C}^n$, a column vector of complex entries. The identity for $+$ (the **zero** vector) is just written 0 .

$\langle u|$ A **bra** is a vector in the **dual** space V^* , i.e. scalar-valued linear maps $V \rightarrow \mathbb{C}$ — a row vector in $\mathbb{C}^n$.

There is a bijective correspondence between $|u\rangle$ and $\langle u|$

$$|u\rangle = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \Leftrightarrow [\bar{u}_1 \cdots \bar{u}_n] = \langle u|$$

A Language: Dirac's notation

Multiplying a **bra** and **ket** gives a ...

$$\text{a bracket: } \langle u | v \rangle \rightsquigarrow \langle u | v \rangle$$

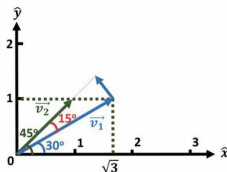
which provides the remaining link in our underlying mathematics:

the **inner product**

$$\text{Notation: } \langle u | v \rangle \equiv \langle u, v \rangle \equiv (|u\rangle, |v\rangle)$$

Recall: what is a inner product in $\mathbb{R}^2$?

From a linear algebra textbook: for $\vec{v}_1 = \sqrt{3}\hat{x} + \hat{y}$ and $\vec{v}_2 = \hat{x} + \hat{y}$, the **inner** (or **dot**, or **scalar**) product is



$$|\vec{v}_1||\vec{v}_2|\cos 15 \approx 2.732$$

i.e. the product of the (magnitudes of) $\vec{v}_2$ by $\vec{v}_1$ projected in the direction of $\vec{v}_2$.

The inner product measures **how much the two vectors have in common after being scaled by their magnitudes**.

Recall: what is a inner product in $\mathbb{R}^2$?

This definition is equivalent to another one which explicitly involves the vectors' coordinates:

$$\vec{v}_1^T \vec{v}_2 = \begin{bmatrix} (v_1)_1 & (v_1)_2 & \dots \end{bmatrix} \begin{bmatrix} (v_2)_1 \\ (v_2)_2 \\ \dots \end{bmatrix}$$

Or, in Dirac's notation in $\mathbb{R}^2$,

$$\langle v_1, v_2 \rangle = \langle v_1 || v_2 \rangle = \langle v_1 | v_2 \rangle$$

Thus,

$$\langle v | v \rangle = v_1^2 + v_2^2 + \dots$$

which entails **positivity**, i.e.

$$\langle v | v \rangle \geq 0 \quad \text{with} \quad \langle v | v \rangle = 0 \quad \text{iff} \quad |v\rangle = 0$$

The underlying maths: Hilbert spaces

Complex, inner-product vector space

A complex vector space with **inner product**

$$\langle - | - \rangle : V \times V \longrightarrow \mathbb{C}$$

such that

$$(1) \quad \langle v | \sum_i \lambda_i \cdot |w_i\rangle \rangle = \sum_i \lambda_i \langle v | w_i \rangle$$

$$(2) \quad \langle v | w \rangle = \overline{\langle w | v \rangle}$$

$$(3) \quad \langle v | v \rangle \geq 0 \quad (\text{with equality iff } |v\rangle = 0)$$

Note: $\langle - | - \rangle$ is **conjugate linear** in the first argument:

$$\langle \sum_i \lambda_i \cdot |w_i\rangle | v \rangle = \sum_i \overline{\lambda_i} \langle w_i | v \rangle$$

and **linear** in the second.

Inner product: examples

In $\mathbb{C}$

$$\langle a + bi | c + di \rangle = \overline{(a + bi)}(c + di)$$

In $\mathbb{C}^n$

$$\langle u | v \rangle = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \underbrace{\begin{bmatrix} \overline{u_1} & \overline{u_2} & \cdots & \overline{u_n} \end{bmatrix}}_{\langle u |} \underbrace{\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}}_{| v \rangle} = \sum_{i=1}^n \overline{u_i} v_i$$

An important note

Defining the dual vector in $\mathbb{C}^n$ in terms of the **conjugate transpose** preserves **positivity** of the internal product.

Indeed, in $\mathbb{R}^n$ positivity holds because the square of a real number is always positive, which is not the case for $\mathbb{C}$. However,

$$\overline{(a + bi)}(a + bi) = (a - bi)(a + bi) = a^2 + b^2 \geq 0$$

Maps in a Hilbert space

Maps between Hilbert spaces are, of course, **linear transformations** which are typically represented by **matrices** whose entries are computed through the inner product:

$$A_{i,j} = \langle i|A|j\rangle$$

Recall we resorted to notation $\langle i|v\rangle$ to represent the coefficient in $|i\rangle$ of vector $|v\rangle$, i.e. its entry in line i .

The dual space

 V^*

If V is a Hilbert space, V^* is the space of **linear maps** from V to $\mathbb{C}$.

Elements of V^* are denoted by

$$\langle u| : V \longrightarrow \mathbb{C}$$

as discussed above, and defined through the **inner product**:

$$\langle u|(|v\rangle) = \langle u|v\rangle$$

In a matricial representation $\langle u|$ is the **Hermitian conjugate**, or **conjugate transpose** of $|u\rangle$,

i.e. the **transpose** of the vector formed by the **complex conjugate** of each element in $|u\rangle$.

The adjoint object

The bijective correspondence between V and V^* is established by $\dagger$:

$$(|u\rangle)^\dagger = \langle u| \quad \text{and} \quad (\langle u|)^\dagger = |u\rangle$$

which is **antilinear**:

$$\left(\sum_i \alpha_i |u_i\rangle \right)^\dagger = \sum_i \overline{\alpha_i} \langle u_i|; \quad \text{and} \quad \left(\sum_i \alpha_i \langle u_i| \right)^\dagger = \sum_i \overline{\alpha_i} |u_i\rangle$$

The adjoint map

The adjoint $U^\dagger$ of a map $U : V \longrightarrow V$ is the unique map satisfying

$$(U^\dagger|w\rangle, |v\rangle) = (|w\rangle, U|v\rangle) \quad \text{or, in a simplified notation}$$
$$\langle U^\dagger w | v \rangle = \langle w | Uv \rangle$$

an equality that is often denoted by the expression

$$\langle u | U | v \rangle$$

Concretely,

$$\langle i | U^\dagger | j \rangle = \overline{\langle j | U | i \rangle}$$

i.e., the matrix representation of $U^\dagger$ is the conjugate transpose of U

Properties

- $(UV)^\dagger = V^\dagger U^\dagger$
- $U^{\dagger\dagger} = U$

Other old friends: Norms and orthogonality

Old friends

- $|v\rangle$ and $|w\rangle$ are **orthogonal** if $\langle v|w\rangle = 0$
- **norm**: $\|v\| = \sqrt{\langle v|v\rangle}$, a nonnegative real number

This norm satisfies the **triangle** equalities $\|v\rangle + |w\rangle\| \leq \|v\rangle\| + \|w\rangle\|$ due to the **Cauchy-Schwarz inequality**:

$$|\langle x|y\rangle| \leq \|x\| \|y\|$$

Other old friends: Norms and orthogonality

Recall

- **normalization**: $\frac{|v\rangle}{||v\rangle|}$
- $|v\rangle$ is a **unit vector** if $||v\rangle| = 1$
- A set of vectors $\{|i\rangle, |j\rangle, \dots, \}$ is **orthonormal** if each $|i\rangle$ is a unit vector and

$$\langle i | j \rangle = \delta_{i,j} = \begin{cases} i = j & \Rightarrow 1 \\ \text{otherwise} & \Rightarrow 0 \end{cases}$$

Other old friends: Bases

Orthonormal basis

A orthonormal basis for a Hilbert space V of dimension n is a set $B = \{|i\rangle, \dots\}$ of n linearly independent elements of V st

- $\langle i|j\rangle = \delta_{i,j}$ for all $|i\rangle, |j\rangle \in B$
- and B **spans** V , i.e. every $|v\rangle$ in V can be written as

$$|v\rangle = \sum_i \alpha_i |i\rangle \quad \text{for some } \alpha_i \in \mathbb{C}$$

Changing representations of a quantum state from one basis to another is a common technique in quantum algorithms.

(cf *why superposition can help you to date the girl/boy of your choice?*)

Example: The Hadamard basis

One of the infinitely many orthonormal bases for a space of dimension 2:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Check e. g.

$$\langle + | - \rangle = \frac{1}{2} \langle (|0\rangle + |1\rangle), (|0\rangle - |1\rangle) \rangle = \frac{1}{2} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0$$

$$\|+\rangle\| = \sqrt{\langle + | + \rangle} = \sqrt{\frac{1}{2} \langle |0\rangle + |1\rangle, |0\rangle + |1\rangle \rangle} = 1$$

Old friends: Bases

If $|v\rangle$ is expressed wrt an orthonormal basis, i.e. $|v\rangle = \sum_i \alpha_i |i\rangle$, then the **amplitude** of $|v\rangle$ wrt $|i\rangle$ satisfies

$$\alpha_i = \langle i|v\rangle$$

because

$$\begin{aligned}\langle i|v\rangle &= \langle i|\sum_j \alpha_j |j\rangle \\ &= \sum_j \alpha_j \langle i|j\rangle \\ &= \sum_j \alpha_j \delta_{i,j} \\ &= \alpha_i\end{aligned}$$

Bases

A basis for $V^\dagger$

If $\{|i\rangle, \dots\}$ is an orthonormal basis for V , then

$$\{\langle i|, \dots\}$$

is an orthonormal basis for $V^\dagger$

Old friends: Bases

Thus, a vector $\langle v|$ in the dual space is expressed wrt an orthonormal basis as

$$\langle v| = \sum_i \overline{\alpha_i} \langle i|$$

where

$$\overline{\alpha_i} = \langle v|\alpha_i\rangle$$

Hilbert spaces

The complete picture

An **Hilbert space** is an inner-product space V st the metric defined by its norm turns V into a **complete metric space**, i.e.any Cauchy sequence

$$|v_1\rangle, |v_2\rangle, \dots$$

$$\forall_{\epsilon>0} \exists_N \forall_{m,n>N} \|v_m - v_n\| \leq \epsilon$$

converges

(i.e. there exists an element $|s\rangle$ in V st $\forall_{\epsilon>0} \exists_N \forall_{n>N} \|s - v_n\| \leq \epsilon$)

The completeness condition is trivial in **finite dimensional** vector spaces